Evaluating the Efficacy of Active Learning Interventions in Undergraduate STEM Education: A Randomized Trial

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ABSTRACT
Background: Active learning strategies have gained prominence as alternatives to traditional didactic lecturing. Howev
Methods: We conducted a two-arm randomized controlled trial involving undergraduate students enrolled in introducto
Results: A total of 184 undergraduate students were recruited (92 in the active learning intervention group, 92 in the tr
Conclusion: Incorporating structured problem-solving and peer discussion significantly improves cognitive retention ar

1. INTRODUCTION
Traditional lecturing methods rely heavily on passive content absorption. Recent meta-analyses indicate that active er

2. METHODS AND PARTICIPANTS

2.1 Participant Selection and Setting

This study was conducted across two university campuses in the United Kingdom between September and Decembe

2.2 Randomization and Study Design

We utilized a parallel-group randomized controlled trial design. Participants were assigned using a computer-generate

1) The Active Learning Group (n = 92): Received 3 hours per week of interactive case studies, think-pair-share activiti

2) The Control Lecture Group (n = 92): Received 3 hours per week of standard didactic slides delivered by the same i

2.3 Primary and Secondary Outcome Measures

The primary outcome was cognitive mastery measured by a standardized comprehensive final examination (0 to 100%

3. RESULTS

3.1 Primary Academic Outcomes

All 184 participants completed the final examination at week 12 with zero attrition. As hypothesized, the active learning
An independent samples t-test confirmed a statistically significant difference between cohorts: $t(182) = 8.84$, $p < 0.001$

3.2 Intervention Duration and Adherence

The complete intervention duration spanned exactly 12 academic weeks. Average attendance in the active learning w

4. DISCUSSION AND CONCLUSION

Our findings corroborate previous cognitive psychology studies demonstrating the superiority of participatory educatio